Deadlocks: what this chapter explains

- Distinguish necessary from sufficient deadlock conditions
- Prove how a global lock order excludes circular waiting
- Apply the Banker's safety test using complete resource vectors

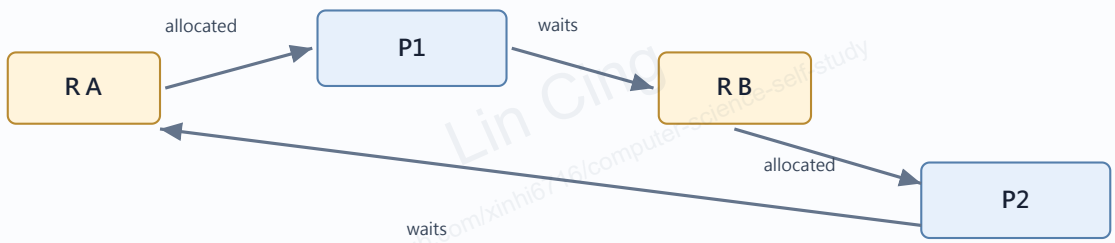
Central question. When can resource waiting become permanent, and which prevention or recovery strategy is justified by available information?

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System model

- A resource type has a defined number of instances. Processes request, use, and release instances; an unsatisfied request may block.
- In a resource-allocation graph, process-to-resource edges represent requests and resource-instance-to-process edges represent allocations. The graph records specified ownership and waiting, not every possible future action.

Single-instance cycle: A → P1 → B → P2 → A



Two single-instance owners wait for each other and cannot be preempted: deadlocked.

Resource state and concrete deadlocks

A resource type groups interchangeable instances. Two printers with incompatible location/function requirements belong to different types. Tables record type totals, free/allocated instances, owners, and each outstanding requester/type. In a deadlocked set every member waits for an event that only a waiting member of the same set can cause, with no permitted outside way to unblock it. The member can be itself: a thread holding a nonrecursive mutex that blocks trying to acquire it again cannot reach the release. This is a one-thread deadlock.

With three CD-RW drives, P1/P2/P3 each holds one and requests one more. Available=0 and none can finish, so all three deadlock. With one printer and one DVD, P1 holds DVD and requests printer; P2 holds printer and requests DVD. Neither can release, so their distinct resource types still deadlock.

Exact graph criteria and a multiple-instance counterexample

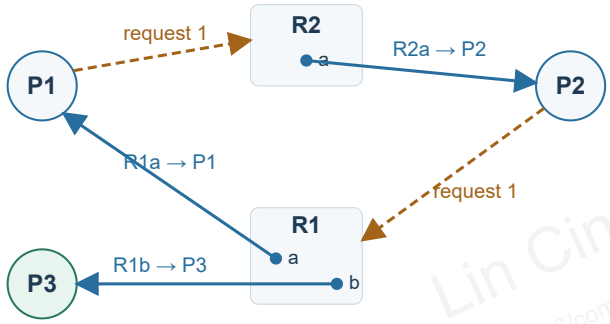
Mutual exclusion makes at least one resource exclusive. Hold-and-wait keeps an allocation while another request waits. No preemption forbids forced reclamation. Circular wait gives a dependency chain returning to its start. Circular wait already implies hold-and-wait; the four conditions are not logically independent.

A process node is a circle, resource type a box, instances dots. Request Pi→Rj becomes assignment Rj→Pi when granted; release removes the assignment. Assume a consistent snapshot: each request waits for one instance, cannot time out or be cancelled, and a blocked owner cannot be preempted or release its holdings early. No cycle implies no resource deadlock. If every resource type in the entire graph has one instance, there is a cycle if and only if there is a deadlock. Even in a mixed system, one particular cycle whose resource types all have one instance is sufficient to prove its participants deadlocked. A cycle involving a multiple-instance type alone does not decide deadlock: another instance holder may finish and release it. Not every deadlock in a mixed system must contain a cycle consisting solely of single-instance resource types.

Why can a cycle still make progress?

R1 has instances R1a/R1b; R2 has R2a. P3 needs nothing more. Each process finishes and returns all holdings after its current request is granted. Available=(0,0).

Initial state: Total=(2,1), Available=(0,0)



Can finish; no more requests

Read arrow direction

Solid: instance → owner

Dashed: waiter → resource type

R1 contains two distinct instances.

Cycle: P1 → R2 → P2 → R1 → P1

P3 holds the other R1 instance.

Let P3 finish to unlock progress.

P3 can release R1b even while P1 and P2 wait on the cycle.

Finish P3, then unblock P2 and P1

Completion order P3 → P2 → P1; vector order (R1,R2)

Event	R1a owner	R1b owner	R2a owner	Available
Initial: P1 and P2 wait	P1	P3	P2	(0, 0)
P3 finishes; returns R1b	P1	free	P2	(1, 0)
Grant R1b to P2	P1	P2	P2	(0, 0)
P2 finishes; returns R1b, R2a	P1	free	free	(1, 1)
Grant R2a to P1	P1	free	P1	(1, 0)
P1 finishes; returns R1a, R2a	free	free	free	(2, 1)

A grant decreases Available; an actual completion returns holdings and increases it. The completion order P3→P2→P1 alone omits the grants to P2 and P1. The figure includes both intermediate deductions.

Detection's Work is a simulation, not live inventory

Vectors use order (R1,R2). Current outstanding requests and existing allocations give:

Process	Allocation	Request	Work before	Work after simulated completion
P3	(1,0)	(0,0)	(0,0)	(1,0)
P2	(0,1)	(1,0)	(1,0)	(1,1)
P1	(1,0)	(0,1)	(1,1)	(2,1)

Check Request≤Work before each step, then set Work←Work+Allocation. This combines granting Request with later returning Allocation+Request, for a net increase of Allocation only. Work may include returns by processes simulated as finished. **The simulation does not execute them; initial Available remains (0,0).**

This test uses **current outstanding Request**, not the Banker's maximum remaining Need. Section 7.6 gives the complete detection algorithm.

Change only P3's next action to obtain deadlock

Keep every allocation and the requests of P1 and P2. Now require P3 to obtain one R2 before it can finish, adding P3→R2.

Initial state	P3 Request	Eligible at Work=(0,0)	Result
Original	(0,0)	P3	Completion order P3→P2→P1
Modified	(0,1)	None	P1, P2, P3 are deadlocked

P1 and P3 now both wait for R2a, held by P2. Both instances of R1 needed by P2 are held by those waiting processes. Every instance is trapped within the waiting set, with no owner able to return one first. This assumes no preemption, cancellation, extra instances, or early release before completion. Merely delaying P3's CPU service does not change resource feasibility in the original case; scheduling starvation and resource deadlock are different questions.

A single-instance lock cycle

Thread A acquires `first` then `second`; B uses the reverse order.

Time	Action	Holdings and waits
t0	A acquires first	A holds first
t1	B acquires second	A holds first; B holds second
t2	A requests second	A holds first and waits for B
t3	B requests first	B holds second and waits for A

Both locks have one instance. Neither owner can proceed or release, and no external holder can help. This interleaving deadlocks. If A finishes before t1, that schedule avoids the cycle, but one successful stress schedule cannot exclude the bad schedule.

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Lock-order proof

- If a thread may request a new lock only above every lock it already holds, each dependency follows an increasing rank. A cycle would require rank(A)<...<rank(A), a contradiction.
- Callbacks, lock upgrades, condition-wait reacquisition, and exceptional paths must obey the same rule. Checking only the normal path is incomplete.

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Policy information and cost

Policy	Information	Guarantee	Cost
Prevention	Restricted request protocol	Removes a necessary condition	Underutilization, refused work, rollback
Avoidance	Trustworthy declared maxima	Every grant preserves a safe sequence	Claims and runtime safety checks
Detection/recovery	Current ownership/request snapshot	Finds and repairs allowed deadlocks	Delay, scanning, lost victim work
Ignore	No general deadlock protocol	No general guarantee	Application timeout or restart

For one R1 and one R2, prevention orders F(R1)=1<F(R2)=2; a holder of R2 cannot request R1. Avoidance uses prior claim edges and refuses a grant that would create a cycle. Detection grants first and scans the wait-for cycle later. Their decision times differ.

Sharing read-only files removes exclusion only for those resources. Acquiring everything initially removes hold-and-wait but can reserve an idle printer until the final stage. Releasing everything before a new request may destroy uncommitted state. Preemption is useful only for restorable resources; it can reclaim a blocked process's allocations and restart when all needed resources become available. Strict ordering removes circular wait.

Policy information and cost (continued)

With injective F:R→N, a cycle would require F(R0)<F(R1)<…<F(Rn)<F(R0), hence F(R0)<F(R0), a contradiction. Sort dynamic account locks by stable ID before acquiring them; from/to parameter order can reverse across transactions. Callback, upgrade, wait reacquisition, and error paths participate in the same proof.

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Claim edges and the complete Banker algorithm

For single instances, a claim Pi→Rj means Pi may request Rj later. A request changes claim to request, then trial assignment. Grant only if the resulting graph remains acyclic. Claims must be known before start, or added only while that process holds no assignments. Missing claims invalidate the guarantee. An adjacency-matrix cycle method costs $O(n^2)$.

For n processes and m resource types, Need=Max−Allocation and comparisons are componentwise. Legal inputs have nonnegative integers and $0 \le \text{Allocation} \le \text{Max} \le \text{Total}$.

```
Work = Available
Finish = [false] * n
while there is unfinished i with Need[i] <= Work:
    Work += Allocation[i]
    Finish[i] = true
safe iff all Finish
```

Once $\text{Need}_i \le \text{Work}$, lending Need_i and receiving $\text{Allocation}_i+\text{Need}_i$ has net effect $\text{Work}-\text{Need}_i+(\text{Allocation}_i+\text{Need}_i)=\text{Work}+\text{Allocation}_i$. It neither manufactures Need nor double-counts the temporary loan. Assume participants finish and return everything once their declared remaining needs are met.

Claim edges and the complete Banker algorithm (continued)

The greedy choice needs no backtracking. If a safe order exists and a different currently feasible Pi is chosen first, move Pi to the front and keep all other processes in their original order. Returning Pi's nonnegative allocation only increases resources faced by every remaining step. If the algorithm stops unfinished despite a supposed safe order, take the first unfinished Pj in that order. Every predecessor has already been removed by the algorithm; current Work contains their allocations and possibly more. Thus $\text{Need}_j \le \text{Work}$, contradicting the absence of a candidate. Completion provides a safe-order certificate; stopping unfinished proves unsafety within this model.

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Resource-request transaction

Validate an m-component nonnegative integer Request_i. If any component exceeds Need_i, reject as exceeding the claim. Otherwise if any component exceeds Available, wait. Negating componentwise $\leq$ means some component is greater, not all components.

Trial Available−=Request, Allocation_i+=Request, Need_i−=Request. Run safety. Commit only if safe; otherwise restore all three arrays and wait. The update is transactional. Typical safety complexity is $O(mn^2)$, at most n scans of n rows with m-component tests.

Counterexample: Total=(2,1), Allocation=(1,0), Need=(1,1), Available=(1,1) is legal. Request=(2,0) exceeds Need and Available in the first component. It must error before trial. Requiring both components to be greater would wrongly allow Available=(−1,1).

Banker T0 and every Work state

Total=(10,5,7), Available=(3,3,2).

Process	Allocation	Max	Need
P0	(0,1,0)	(7,5,3)	(7,4,3)
P1	(2,0,0)	(3,2,2)	(1,2,2)
P2	(3,0,2)	(9,0,2)	(6,0,0)
P3	(2,1,1)	(2,2,2)	(0,1,1)
P4	(0,0,2)	(4,3,3)	(4,3,1)

Each next Need fits. Work proceeds (3,3,2)→P1 (5,3,2)→P3 (7,4,3)→P4 (7,4,5)→P0 (7,5,5)→P2 (10,5,7). Thus ⟨P1, P3, P4, P0, P2⟩ is safe.

P1 requests (1,0,2), within Need(1,2,2) and Available. Trial gives Available=(2,3,0), Allocation1=(3,0,2), Need1=(0,2,0). The same order yields Work (2,3,0)→(5,3,2)→(7,4,3)→(7,4,5)→(7,5,5)→(10,5,7), so grant.

After that grant, P0 requests (0,2,0). Although available, its trial leaves Work=(2,1,0). No remaining Need fits, including P0's updated Need=(7,2,3), so rollback and wait. Unsafe means the declared worst-case completion guarantee is missing. Voluntary early release or smaller actual demands can still avoid a real deadlock.

Five resource types: find the smallest safe stock

Four processes share five non-preemptible, reusable resource types. Once a process receives its maximum remaining demand, it finishes in finite time and releases all holdings. Vectors are ordered by resources A, B, C, D, E.

Process	Allocation	Max	Need = Max − Allocation
P1	(1,0,2,1,1)	(1,1,5,1,2)	(0,1,3,0,1)
P2	(2,0,1,1,0)	(5,2,2,1,0)	(3,2,1,0,0)
P3	(1,1,0,1,0)	(3,1,3,1,0)	(2,0,3,0,0)
P4	(1,1,1,1,0)	(1,1,2,2,1)	(0,0,1,1,1)

Available = (0,0,x,1,1), where x is a nonnegative integer. Each x corresponds to Total = ΣAllocation + Available = (5,2,4+x,5,2). Find the smallest x giving a safe state. Different x values specify different candidate totals, not a changing inventory in one fixed system.

Follow Work to find the smallest x

x = 2: check Need ≤ Work at each step

Finishes	Old Work	Its Need		New Work
P4	(0,0,2,1,1)	(0,0,1,1,1)	→	(1,1,3,2,1)
P1	(1,1,3,2,1)	(0,1,3,0,1)	→	(2,1,5,3,2)
P3	(2,1,5,3,2)	(2,0,3,0,0)	→	(3,2,5,4,2)
P2	(3,2,5,4,2)	(3,2,1,0,0)	→	(5,2,6,5,2)

New Work = old Work + the process's Allocation

At x=0 no Need fits the initial Work. At x=1 only P4 can finish first; after its simulated release Work=(1,1,2,2,1). P1 lacks C, P2 lacks A and B, and P3 lacks A and C, so no next process fits.

At x=2, check Need ≤ Work along the diagram to obtain the safe sequence ⟨P4,P1,P3,P2⟩. Thus the smallest integer is **x=2**. Work is a simulation ledger: passing the check does not mean processes have finished or that real Available has become the final Work.

One resource type: a sufficient safety bound

There are m identical reusable instances and a finite set of n processes. Process i declares $1 \leq Max_i \leq m$. Its holdings are not preempted before completion; it blocks on a resource request only when insufficient instances are available, and completes in finite time and releases all holdings once its required resources are provided.

For unfinished processes, set $Need_i = Max_i - Allocation_i$. If

$$\sum_{i=1}^{n} Max_i < m + n,$$

every legal allocation has an unfinished process with $Need_i \leq$ Available. Completing such a process and repeating gives a safe completion sequence.

For m=10 and four processes each with Max=3, ΣMax=12 < 14, so every legal allocation is safe in this model. The condition is **sufficient, not necessary**: with five processes each having Max=3, ΣMax=15 is not less than m+n=15, yet the state in which nobody holds any instances and Available=10 is safe.

Why does every round have a candidate?

Let v be the current number of available instances. Suppose none of the n unfinished processes fits its maximum remaining demand. Then each integer $Need_i > v$ implies $Need_i \geq v+1$. Since $\Sigma Allocation_i = m - v$,

$$\sum_i Max_i = \sum_i Allocation_i + \sum_i Need_i \geq (m - v) + n(v + 1) = m + n + (n - 1)v \geq m + n,$$

contradicting the bound. A candidate therefore exists. This argument covers v>0 as well as the fully allocated case.

On completion, subtract its $Max_i \geq 1$ and reduce the process count by one; the strict inequality still holds for the remaining processes. Repeating yields a safe sequence. This is a resource-model existence guarantee: actual progress still requires sensible granting and scheduling, and the inequality alone cannot prevent starvation under an unfair scheduler.

Same free resources, different safety decisions

Use resource types A and B, Total=(4,3), and initial state S with Available=(1,1). Every vector is ordered A then B. The two trials below start independently from S.

Process	Allocation	Max	Need=Max−Allocation
P0	(1,0)	(3,2)	(2,2)
P1	(1,1)	(2,2)	(1,1)
P2	(1,1)	(3,2)	(2,1)

Allocated totals (3,2), so adding Available recovers Total. Each row satisfies Allocation+Need=Max and Max≤Total. Resources are nonpreemptible and returned together at completion. A process finishes in finite time once its actual demand is supplied.

S is safe: P1 fits Work=(1,1), then returns its allocation to give (2,2). P0 gives (3,2), then P2 gives (4,3). This is a simulated safety certificate, not a required CPU schedule.

Who receives the request changes the result

Each trial starts from S and requests (1,0). Both requests fit the requester's Need and Available. The second trial does not follow the first.

Same starting state and request; different safety outcomes

S: Available=(1,1); each branch starts independently from S

P0 requests (1,0)

Trial: Available=(0,1)

Allocation0=(2, 0)

Need0=(1, 2)

No Need fits (0,1)

unsafe: roll back all three updates

P1 requests (1,0)

Trial: Available=(0,1)

Allocation1=(2, 1)

Need1=(0, 1)

Simulated order: P1 → P0 → P2

safe: commit the trial state

Both trials leave (0,1), but ownership and remaining claims differ.
A safety order is a simulated certificate, not an executed schedule.

Availability alone is insufficient. Giving A to P1 leaves it needing only one B to finish. Giving the same A to P0 leaves nobody able to finish against its maximum remaining claim.

Reject P0: identify every failing component

Call P0's trial state U. Available=(0,1); only P0 changes, to Allocation0=(2,0), Need0=(1,2). Initialize Work=(0,1) and all Finish entries false.

Candidate	Trial Need	Failing component
P0	(1,2)	A: 1>0; B: 2>1
P1	(1,1)	A: 1>0
P2	(2,1)	A: 2>0

No candidate exists, so safety fails immediately. Restore S: add (1,0) to Available, subtract it from Allocation0, and add it to Need0. Leave other rows and Max unchanged.

Restoring only Available double-counts A. Available+ΣAllocation would be (5,3), exceeding Total=(4,3). Other requests must not observe a partial transaction. Protect check/commit with a lock, or validate the snapshot version at commit and retry after interference.

Grant P1: safety does not execute processes

Restart at S. Trial P1's (1,0) gives Available=(0,1), Allocation1=(2,1), Need1=(0,1). This table updates only simulated Work and Finish.

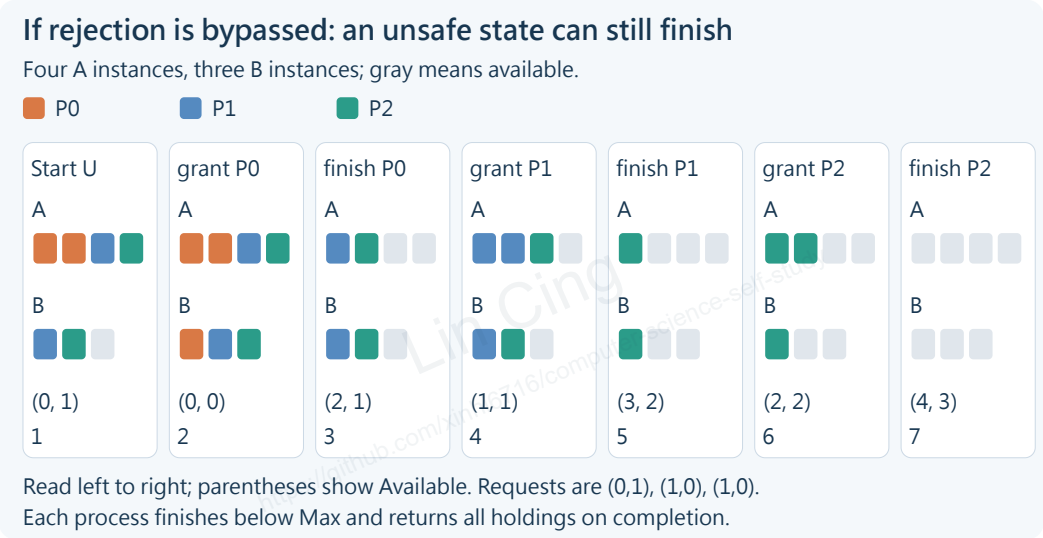
Simulated finish	Need≤old Work	Return Allocation	New Work	Finish: P0/P1/P2
Initial	—	—	(0,1)	F/F/F
P1	(0,1)≤(0,1)	(2,1)	(2,2)	F/T/F
P0	(2,2)≤(2,2)	(1,0)	(3,2)	T/T/F
P2	(2,1)≤(3,2)	(1,1)	(4,3)	T/T/T

Commit the **trial state**, whose Available remains (0,1). Do not copy final Work=(4,3) into Available or treat simulated Finish as actual termination. Real instances return only after actual completion.

Work adds Allocation because granting Need and later returning Allocation+Need has net change −Need+(Allocation+Need)=Allocation. Need creates no resources.

Unsafe does not mean deadlocked: an actual completion path

Suppose P0's rejected trial U is committed anyway. Max bounds demand; it need not be exhausted.



Requests (0,1), (1,0), (1,0) fit the claims in U. This is **one** successful future, not a worst-case guarantee.

The same U also permits deadlock

At U, suppose P0/P1/P2 instead request their entire remaining Need: (1,2), (1,1), (2,1). Each request respects Max, but Available=(0,1) satisfies none. All block while retaining A. With no cancellation, preemption, or early return, this is deadlock.

The two futures share Allocation, Max and Available; their actual requests differ. Safety uses maximum remaining Need to certify a conservative completion order. Detection uses current outstanding Request. Do not silently turn a possible future Need into a present Request.

An unsafe state has no safe sequence, yet can have a successful actual execution. Safe sequence is defined against maximum claims. An observed completion order need not be a safe sequence in that technical sense.

Detection reduction and recovery states

For single instances, replace Pi→Rq→Pj by Pi→Pj. A cycle is deadlock exactly under the exclusive single-instance assumptions. A cycle through a multiple-instance type needs allocation/request analysis; a concrete cycle containing only single-instance types still proves its members deadlocked under the stated blocking assumptions.

```
Work = Available
Finish[i] = (Allocation[i] == zero_vector)
while there is i with not Finish[i] and Request[i] <= Work:
    Work += Allocation[i]
    Finish[i] = true
remaining false holders form the detected blocked set
```

Detection uses current Request, not maximum Need. In the optimistic reduction model, a process whose current request is met completes in finite time, returns all its allocation, and issues no additional requests before completion. Initializing zero-allocation processes to true means they hold no resources needed by others; it does not assert that they actually finished or are not waiting. New requests need another snapshot. A consistent snapshot is required; reduction costs $O(mn^2)$.

Detection: reduce the baseline snapshot

Available=(0,0,0). The complete base snapshot is:

Process	Allocation	Request
P0	(0,1,0)	(0,0,0)
P1	(2,0,0)	(2,0,2)
P2	(3,0,3)	(0,0,0)
P3	(2,1,1)	(1,0,0)
P4	(0,0,2)	(0,0,2)

Work transitions P0→(0,1,0), P2→(3,1,3), P3→(5,2,4), P1→(7,2,4), P4→(7,2,6). All Finish become true: under the stated optimistic model, this snapshot admits a simulated order that clears the current requests. This does not guarantee that future real requests cannot cause deadlock.

Detection: alter one request

Change only Request2 to (0,0,1). Simulating P0 gives Work=(0,1,0); no other Request fits, so Finish=(true, false, false, false, false) and the unresolved holders are {P1, P2, P3, P4}. With no cancellation, preemption or early return, they form a deadlocked set in this model. Simulation does not actually complete P0 or change Available. If P0 has not finished, aborting P1 and safely reclaiming Allocation1=(2,0,0) changes actual Available from (0,0,0) to (2,0,0).

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Recovery: rerun reduction after termination

Restart reduction with Work=Available, excluding the terminated P1:

Simulated completion	Current Request	Work after returning Allocation
Initial; P1 terminated	—	(2,0,0)
P0	(0,0,0)	(2,1,0)
P3	(1,0,0)	(4,2,1)
P2	(0,0,1)	(7,2,4)
P4	(0,0,2)	(7,2,6)

Check Request≤Work before each row and add only the existing Allocation. Assuming each survivor finishes in finite time without new requests after its current Request is granted, this certifies a possible completion order; actual Available rises only when they really return resources. Starting at actual Available=(2,1,0) is valid only if P0 has already completed outside the simulation.

Frequent scans reduce detection delay and can identify the latest requester, but cost more. Periodic scans can see several cycles together. A snapshot does not guarantee every future state.

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Recovery unit and victim fairness

Abort all deadlocked processes to break every cycle at the cost of all unfinished work, or abort one, reclaim its resources, rerun detection, and stop once survivors can progress. Victim policy considers priority, interactivity, executed/remaining time, held/needed resources, reclamation difficulty, and how many aborts are needed.

Files and devices can retain partial external effects. Recovery needs undo, journals, checkpoints, or device reset, not merely unlocked mutexes. Preemption must specify victim, consistent rollback point, and protection against repeated victimization.

T2 has checkpoint cost 2 and rollback count 0; T4 cost 3/count 0. Choose T2, roll back and release 2 resources. Rerun detection; if T4 now completes, stop aborting. Increment T2's rollback count. A cost such as execution_i+k×rollback_count_i penalizes repeated selection, but a formal finite bound also needs workload assumptions. A limit on consecutive selections alone does not ensure fairness: A and B can alternate forever.

A single-instance P1/P2 wait cycle is deadlock. Separately, a ready P3 selected zero times while P4 completes five admissions is evidence of possible starvation; only an unbounded execution or policy argument proves no finite bound. No fabricated wait cycle is needed. Ordering can remove deadlock while a CPU policy still starves the lock holder.

Distinguish the evidence for each conclusion

Observation	Conclusion	Additional evidence
Four conditions	Deadlock possible	No remaining completion transition
Single-instance graph no cycle	No deadlock in snapshot	Validated graph
Single-instance cycle	Deadlock	All cycle resources truly single-instance
Multi-instance cycle	Deadlock possible	Availability/reduction
Unsafe	No worst-case avoidance guarantee	Actual current requests
One long wait while others finish	Possible starvation	Admission policy and waiting bound

Chapter conclusions

- Necessary conditions alone do not establish every deadlock claim
- Avoidance depends on valid maximum claims
- Detection, recovery, and data consistency must be designed together

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz, Galvin and Gagne, Operating System Concepts, 9th ed., Chapter 7.

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